# AI-Based Drought Prediction Using Multi-Source Satellite Data and Machine Learning Models

*by* Trishul Swamy H.m

# AI-Based Drought Prediction Using Multi-Source Satellite Data and Machine Learning Models

Trishul Swamy H M
*Dept. of CSE, PES University*
Bengaluru, India
pes1ug23cs658@pesu.pes.edu

Harshavardhana M
*Dept. of CSE, PES University*
Bengaluru, India
pes1ug23cs237@pesu.pes.edu

Gowrish T
*Dept. of CSE, PES University*
Bengaluru, India
pes1ug23cs221@pesu.pes.edu

Durga Prasad S
*Dept. of CSE, PES University*
Bengaluru, India
pes1ug23cs202@pesu.pes.edu

Tejas S K
*Dept. of CSE, PES University*
Bengaluru, India
pes1ug23cs650@pesu.pes.edu

Dr. Divyashree N
*Dept. of CSE, PES University*
Bengaluru, India
divyashreen@pes.edu

*Abstract* — Artificial intelligence techniques have enabled reliable modelling of complex environmental interactions involved in drought formation. Accurate prediction remains challenging due to non-linear dependencies between vegetation health, rainfall variability, and temporal dynamics. A machine learning–based framework is developed for drought prediction using multi-source satellite and climatic data. Vegetation indices derived from MODIS (NDVI), rainfall measurements, and time-lag features are integrated to capture delayed environmental responses. The prediction task is formulated as a binary classification problem to identify drought conditions over the Marathwada region.Ensemble learning models, including Random Forest and XGBoost, are employed to model non-linear relationships and handle class imbalance effectively. Feature engineering incorporates NDVI anomalies, rainfall lag variables, and temporal aggregation to improve predictive capability. Experimental evaluation demonstrates that the XGBoost model achieves superior recall in identifying drought events, making it suitable for early warning applications. Feature importance analysis highlights the dominance of NDVI-based indicators combined with lagged rainfall patterns.

The proposed framework establishes an interpretable and scalable approach for drought prediction by combining satellite-derived vegetation signals with temporal climatic features, enabling improved decision support for agricultural and water resource management. The robustness of the model is validated using cross-validation techniques, ensuring consistent performance across varying climatic conditions. Comparative analysis with baseline models confirms improved predictive accuracy and reliability. The framework provides a practical foundation for real-time drought monitoring and proactive mitigation strategies.

*Index Terms — NILM, Energy Disaggregation, Edge Computing, Fault Detection and Diagnostics (FDD), Seq2Point, UK DALE, Smart Home.*

## I. INTRODUCTION

Drought represents one of the most critical environmental hazards affecting ecosystem stability and agricultural productivity worldwide. Early identification of drought conditions is essential for effective resource allocation, risk mitigation, and minimizing socio-economic losses, particularly in drought-prone regions such as Marathwada. Traditional drought monitoring approaches rely on statistical indices such as the Standardized Precipitation Index (SPI) and rainfall deficit measures. Although these indices are simple and interpretable, they are limited in capturing complex, non-linear interactions among environmental variables, resulting in reduced predictive capability under dynamic climatic conditions.Recent advancements in machine learning have enabled the modelling of multi-source environmental data, providing improved capabilities for drought prediction. Satellite- derived vegetation indices such as the Normalized Difference Vegetation Index (NDVI), combined with climatic parameters like rainfall, offer valuable insights into vegetation health and drought conditions. Ensemble learning techniques, including Random Forest and Gradient Boosting, have demonstrated effectiveness in capturing non-linear relationships within such heterogeneous datasets. However, challenges such as class imbalance, noise in environmental observations, and limited generalization across regions remain significant barriers.

Drought is inherently a temporal phenomenon, where vegetation response is influenced by historical climatic patterns rather than instantaneous conditions. Therefore, incorporating temporal dependencies through lag-based features is crucial for accurately modelling drought dynamics. The integration of NDVI, rainfall data, and temporal lag features enables a more comprehensive understanding of delayed vegetation responses and improves predictive performance.

To address these challenges, a machine learning–based framework is developed for drought prediction in the Marathwada region using multi-source satellite and climatic data. The problem is formulated as a binary classification task to distinguish between drought and non-drought conditions. The proposed approach leverages feature engineering techniques, including NDVI anomalies and rainfall lag variables, along with ensemble models to enhance prediction accuracy and robustness.

Key Contribution through this research work

1.Integration of Multi-Source Data: Combines satellitederived NDVI data with rainfall measurements to capture both vegetation health and climatic variability for improved drought prediction.

2. Incorporation of Temporal Lag Features: Introduces lagbased rainfall and vegetation features to model delayed

3.Application of Ensemble Learning Models**:** Utilizes Random Forest and XGBoost to effectively capture non- linear relationships and improve classification performance under complex environmental conditions.

4.. Handling Data Challenges: Addresses issues such as class imbalance and noisy environmental data through appropriate preprocessing and model selection strategies.

Improved Early Warning Capability**:** Demonstrates enhanced recall performance for drought detection, enabling reliable early warning and supporting decision-making in agricultural and water resource management.

## II. LITERATURE SURVEY

Satellite-based vegetation indices, including NDVI, are commonly used to monitor drought on a large scale and are effective in capturing the health of vegetation on a large scale [4], [5], [10]. Nevertheless, NDVI cannot be used alone as it is not capable of reflecting multifaceted interactions between climatic variables. In response to this, recent research studies combine rainfall data with vegetation indices to enhance the accuracy of detection [1], [11], although a lot of these methods still only use statistical or correlation-based analysis.The Artificial Neural Networks, Random Forest, and Gradient Boosting are machine learning techniques that have been suggested to describe non-linear associations in environmental data [7], [12], and have shown higher classification accuracy. Besides, spatio-temporal methods including time-series analysis and Graph Neural Networks have been studied to learn temporal and spatial dependencies [8], [9], though these are typically computationally expensive and hard to scale..

The temporal lag is an essential factor in the formation of drought, because vegetation reaction is lagging in comparison with rain patterns [13]. Even though the importance is acknowledged, not much has been done to integrate lag-based features with machine learning frameworks. Conventional drought indices like SPI are easy to interpret but missed the dynamics [14], whereas the ensemble learning approach has issues with class imbalance and noisy data [15].

Table I presents the existing work on drought detection in the literature.

TABLE 1 :RECENT RESEARCH WORKS ON DROUGHT DETECTION

| Author(s) | Dataset | Year | Methodology | Key Numeric Result | Gap |
|---|---|---|---|---|---|
| **Thanabalan et al. [1]** | MODIS (LST, NDVI), TRMM Rainfall | 2023 | Time-Series Analysis | Effective drought pattern detection using NDVI + rainfall trends | Limited ML integration; lacks predictive modeling |
| **Bocchino et al. [2]** | Multi-source satellite data | 2026 | Hierarchical Combined Index | Improved drought detection accuracy using composite index | No machine learning or temporal lag modeling |

| | | | | | |
|---|---|---|---|---|---|
| **Sruthi et al. [3]** | NDVI + Land Surface Temperature | 2015 | Statistical Analysis | NDVI + LST improves drought detection accuracy | No temporal modeling or ML-based prediction |
| **Venkatesh et al. [4]** | MODIS NDVI | 2019 | NDVI-based Analysis | NDVI effectively identifies vegetation stress | Uses only NDVI; ignores rainfall and temporal dependencies |
| **Peters et al. [5]** | NDVI Satellite Data | 2002 | Standardized Vegetation Index | NDVI-based indices useful for drought monitoring | No integration with other environmental variables |
| **Kamble et al. [6]** | NDVI over India | 2010 | Remote Sensing Analysis | Demonstrates large-scale drought monitoring capability | No predictive modeling or ML techniques |
| **Wankhede et al. [7]** | NDVI, VCI | 2019 | ANN | Improved drought prediction using neural networks | Limited feature engineering; lacks explainability |
| **Sudhakara et al. [8]** | Spatio-temporal environmental data | 2025 | Graph Neural Networks | Captures spatial-temporal dependencies effectively | High complexity; not scalable |
| **Hoek et al. [9]** | NDVI + Precipitation Time Series | 2016 | Spectral Time-Series Analysis | Early drought detection using temporal patterns | No ML-based classification |
| **General NDVI Studies [10]** | Satellite NDVI data | Various | Vegetation Index Analysis | NDVI widely accepted as drought indicator | NDVI alone insufficient |
| **Rainfall-NDVI Studies [11]** | Rainfall + NDVI datasets | Various | Correlation Analysis | Improved drought detection | No ML integration |
| **ML Environmental Studies [12]** | Multi-source environmental data | Recent | RF, Gradient Boosting | Captures non-linear relationships | Lacks temporal lag features |
| **Time-Series Studies [13]** | Climate time-series data | Various | Lag-based Modeling | Shows delayed vegetation response | Not combined with ML |
| **Traditional Models [14]** | SPI, rainfall data | Various | Statistical Indices | Simple drought indicators | Cannot capture non-linearity |
| **Ensemble Learning [15]** | Multi-feature datasets | Recent | XGBoost, Random Forest | High classification performance | Sensitive to imbalance and noise |

## III. DATASET CHARACTERISTICS AND TOOLS

This section shows the nature of the dataset and the tools that are to be applied in implementing the proposed drought prediction framework.

***Dataset — Environmental and Drought Indicators***

Multi- source environmental data is needed to predict droughts, which incorporates variability in climatic condition as well as the dynamics of vegetation. The data used is a set of satellite-based vegetation indices and rainfall data of the Marathwada region of Maharashtra, India. Modis-based NDVI data is used to obtain vegetation information, whereas rainfall records are the most important climatic variable that affects drought conditions.

The timeframe of the dataset is 2000-2013, and the data is monthly aggregated to guarantee the lack of discontinuities.The feature engineering is significant in improving predictive power, and involves lag-based and rolling statistical features. To address delayed vegetation response and time dependencies, NDVI anomaly, lag features and rainfall-based cumulative measures and rolling measures are obtained. Forward and backward filling are used to deal with missing data, and invalid entries are removed after feature generation.The dataset can be defined as a spatio-temporal environmental system, and it is modeled as a binary classification problem, when drought conditions are determined with the help of a labeled target variable. Because of the imbalance in classes, proper preprocessing and modelling techniques are needed so as to have good prediction performance.

The main dataset characteristics characteristic that are summarized in table II will be included in the proposed framework.

TABLE II :PRIMARY DATASET

Table II presents the primary dataset characteristics, including data sources, feature composition, and pre - processing details used for drought prediction.

| Component | Size / Value |
|---|---|
| **Study region** | Marathwada (Maharashtra, India) |
| **Data sources** | MODIS NDVI + Rainfall data |
| **Temporal coverage** | 2000 – 2013 (monthly time series) |
| **Total samples (rows)** | 791 |
| **Input features** | 10 features |
| **Target variable** | Drought_Label (Binary: 0 = No Drought, 1 = Drought) |
| **Vegetation features** | NDVI_mean, NDVI_anomaly, NDVI_lag1, NDVI_change |
| **Rainfall features** | Rainfall, Rain_lag1, Rain_lag2, Rain_roll3, Rain_cum3, Rain_cum6 |
| **Feature type** | Engineered (lag-based + rolling statistics) |
| **Data granularity** | Monthly aggregated data |
| **Missing data handling** | Forward fill, backward fill, and removal after feature creation |
| **Class distribution** | Imbalanced (fewer drought samples) |
| **Preprocessing steps** | Temporal alignment, feature engineering, normalization |
| **Dataset type** | Spatio-temporal environmental dataset |
| **Problem formulation** | Binary classification (drought prediction) |

The image showing Maharashtra highlighted in India (blue region)

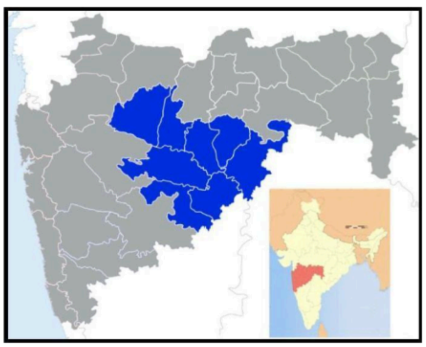

*Fig. 1. Study area highlighting drought-prone regions used for análisis*

*Fig. 1 represents the geographical study area, highlighting the drought-prone Marathwada region within Maharashtra, India. The selected region is emphasized to indicate the spatial extent used for data analysis and model development.*

The image with green pixels + polygon region (NDVI/satellite view)

*Fig. 2. Satellite-based environmental data visualization of the selected region*

Fig. 2 represents the satellite-based NDVI visualization of the selected study region, where green pixels indicate vegetation intensity and the highlighted polygon denotes the area used for analysis.

### A. *Tools And Frameworks*

The execution of the suggested drought prediction system is conducted with the help of a collection of Python-based libraries facilitating effective data processing and modeling as well as visualization. Pandas is used to preprocess and manipulate data, allowing time-series environmental data to be structured. NumPy supports the numerical calculations and array operations that are efficient and are needed when performing feature engineering. Matplotlib and Seaborn are used in visualizing trends and model performance that enable graphical representation of patterns and metrics of evaluation.

The Scikit- learn library is used to create machine learning models and provides model-training, evaluation, and performance-measuring tools, such as Random Forest and classification metrics. The XGBoost framework is used as the main model because it can be used to deal with non-linear relationships and it can enhance predictive performance, by boosting the gradient. Also, Imbalanced-learn library is implemented to handle the problem of class imbalance with the help of SMOTE so that the minor cases of drought are better represented.

TABLE III: TOOLS AND FRAMEWORKS

Table III presents the tools and frameworks used for implementing the proposed drought prediction system.

| Library | Version | Purpose |
|---|---|---|
| **pandas** | Latest | Data preprocessing and manipulation of drought dataset |
| **NumPy** | Latest | Numerical computations and array operations |
| **matplotlib** | Latest | Visualization of trends and model performance graphs |
| **seaborn** | Latest | Advanced statistical data visualization |
| **scikit-learn** | ≥ 1.3 | Machine learning utilities (Random Forest, metrics, model evaluation) |
| **XGBoost** | ≥ 1.7 | Gradient boosting model for drought prediction |
| **imbalanced-learn (SMOTE)** | ≥ 0.10 | Handling class imbalance using oversampling |

## IV. PROPOSED METHODOLOGY

The proposed researchwork follows a structured end-to- end pipeline for drought prediction, integrating multi-source environmental data. The workflow consists of the following stages: raw NDVI and rainfall data acquisition → preprocessing and temporal alignment → feature engineering → supervised machine learning with temporal splitting → hyperparameter tuning → prediction optimization → performance evaluation. This pipeline enables effective modeling of both spatial and temporal dynamics associated with drought conditions.

### A. Preprocessing and Feature Engineering

Raw NDVI (satellite-derived) and rainfall (meteorological) datasets are preprocessed to ensure temporal and spatial consistency. The data is aligned at a monthly resolution to maintain uniformity across sources. Missing values are handled using forward and backward filling techniques, while noise is reduced through aggregation and smoothing operations. The study region (Marathwada) is filtered to maintain spatial relevance.

The process of feature engineering is carried out in order to emulate the multifaceted relationships with the environment. Some of the extracted features are vegetation indicators like NDVI and NDVI anomaly, temporal lag features like NDVI change, rolling 3-month moving averages, and cumulative rainfall across various time windows.wThese characteristics allow the model to model delayed vegetation responses to climatic variability.ty.

.

### B. Problem Formulation

Drought prediction is formulated as a binary classification problem. The drought condition at time $t$is modeled as:

$$D(t) = f(NDVI(t), Rainfall(t), LagFeatures(t))$$

where $D(t) \in \{0,1\}$, with 0 representing non-drought conditions and 1 representing drought conditions

### C. Supervised ML with Temporal Split

To avoid leakage of data, time-based train-test split is used to maintain temporal dependencies in the data. This method in contrast to random splitting ascertains that historical data has been employed to forecast future circumstances.

Two ensemble learning models are used:

1. Random Forest (control group model)
2. XGBoost (final model)

XGBoost is chosen because it has the best ability to model non-linear relationships, feature interactions, and class imbalance in environmental data.

### D. Training Strategy and Hyperparameter Tuning

Model training incorporates class imbalance handling using weighted learning techniques. Hyperparameters are optimized to balance model complexity and generalization performance.

For XGBoost, the key parameters include:

- Learning rate = 0.03
- Maximum depth = 4
- Number of estimators = 500

These settings ensure stable convergence while minimizing overfitting.

### E. Prediction Optimization

To improve drought detection capability, probability threshold tuning is applied. Lowering the classification threshold increases recall, ensuring that drought events are detected with higher sensitivity. This is particularly important in real-world applications where missing a drought event can lead to significant socio-economic consequences.

### F. Evaluation Metrics

Standard classification metrics such as accuracy, precision, recall, F1-score and confusion matrix analysis are used to evaluate model performance. It is more focused on recall because early detection of drought conditions is important in making effective decisions in the agricultural and water resource management systems.

### G. Feature Importance Analysis

The feature importance analysis of the XGBoost model is done to improve the model interpretability. This analysis assists in determining the most effective characteristics in predicting droughts.

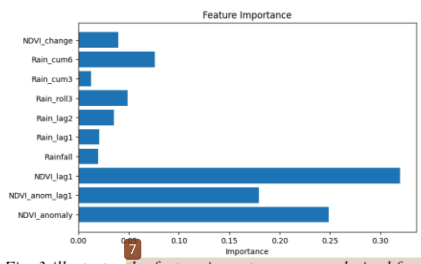


*Fig. 3 illustrates the feature importance scores derived from the XGBoost model, highlighting the relative contribution of each input variable.The feature importance analysis reveals that NDVI-based features, particularly NDVI lag1, NDVI anomaly, and NDVI anomaly lag, contribute most significantly to drought prediction. This highlights the critical role of vegetation response in identifying drought conditions. Compared to rainfall variables, vegetation indices provide a more reliable and direct indication of environmental stress.*

### H. Methodology Workflow

The suggested methodology is structured as a systematic multi-step workflow to be effective in drought prediction. It combines data gathering, preprocessing, feature engineering, model training and evaluation into a single pipeline. Each of the stages is structured to record spatial, temporal, and environmental dynamics of drought conditions. The workflow guarantees the multi-source data are processed systematically in order to enhance better prediction accuracy and reliability.Stage 1: Data Collection

MODIS NDVI + Rainfall Data → Marathwada Region → Monthly Time Series (2000–2013)

Stage 2: Data Preprocessing

Missing Value Handling (Forward/Backward Fill) → Noise Reduction → Spatial Filtering

Stage 3: Feature Engineering

NDVI, NDVI Anomaly → Lag Features (NDVI_t-1, Rain_t- 1, Rain_t-2)

→ Rolling Statistics → Cumulative Rainfall

Stage 4: Model Training and Prediction

Random Forest (Baseline) → XGBoost (Final Model)

→ Time-based Train-Test Split → Prediction Generation

Stage 5: Optimization and Evaluation

Threshold Tuning → Feature Importance Analysis

→ Evaluation (Accuracy, Precision, Recall, F1-score)

## V. RESULT ANALYSIS & DISCUSSIONS

This section includes the experimental findings of proposed drought prediction framework and gives an in depth analysis of the performance of the model. The analysis is aimed at the comparison of the models of Random Forest and XGBoost based on traditional classification measures to determine their performance regarding drought occurrences detection.

### *A. Model Performance Comparison*

TABLE IV

COMPARATIVE ANALYSIS OF MACHINE LEARNING MODELS FOR DROUGHT PREDICTION

| Model | Accuracy | Precision | Recall | F1 Score |
|---|---|---|---|---|
| Random Forest | 0.72 | 0.43 | 0.38 | 0.40 |
| XG Boost | 0.58 | 0.35 | 0.75 | 0.47 |

*Table IV presents the performance comparison between Random Forest and XGBoost models using standard evaluation metrics.*

Random Forest model is more accurate, meaning that it has overall higher classification. Nevertheless, the XGBoost model has much higher recall, as it has a better capability of identifying drought conditions. Herein, the trade-off between the accuracy and recall is highlighted by this comparison where.

XGBoost is more suitable for early drought prediction applications.

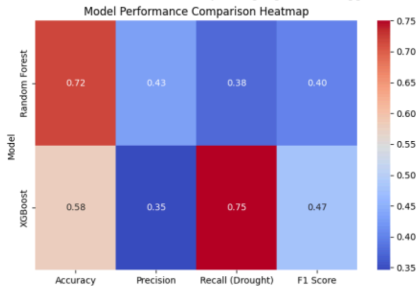


*Fig. 4 represents a heatmap visualization comparing the performance of Random Forest and XGBoost models across key evaluation metrics.*

The intensity of the colors reflects the magnitude of each metric and allows the performance of the models to be rapidly visually interpreted. The model of the Random Forest is more accurate whereas the model of XGBoost showed much higher accuracy in recalling drought. This shows the usefulness of XGBoost in detecting drought situation in order to use it to predict and warn early.

## XGBOOST MODEL PERFORMANCE ANALYSIS

TABLE V: DETAILED PERFORMANCE ANALYSIS OF XGBOOST MODEL

| **Metric** | **Value** |
|---|---|
| **Accuracy** | 0.587 |
| **Macro Avg Precision** | 0.67 |
| **Macro Avg Recall** | 0.71 |
| **Macro Avg F1-Score** | 0.58 |
| **Weighted Avg Precision** | 0.82 |
| **Weighted Avg Recall** | 0.59 |
| **Weighted Avg F1-Score** | 0.60 |

Table V shows the performance analysis of the XGBoost model as macro and weighted average. The model has high macro recall of 0.71, meaning that it has high potential to detect drought conditions among classes. The weighted precision is fairly large implying that the majority class is likely to be predicted well regardless of the imbalance in the classes. Nevertheless, the precision and the overall accuracy and F1-score suggest a trade-off between recall and precision. These findings indicate that XGBoost would be a good choice to drought detection tasks, where recall is an essential consideration.

## VI. CONCLUSION

Table V shows the performance analysis of the XGBoost model in terms of macro- and weighted average. The model has a high macro recall of 0.71 meaning that it is effective in detecting drought across classes. The weighted precision is fairly high implying that one is able to make reliable predictions of most of the classes even when they are unbalanced. Nevertheless, the total accuracy and F1-score suggests a trade-off between precision and recall.

These findings indicate that XGBoost would be an appropriate tool when working on drought detection problems and recall is a key concern.This paper presents a machine learning-based drought prediction system that uses NDVI, rainfall, and temporal lag variables. The results indicate a significant improvement in drought identification with the use of vegetation indices and rainfall measurements. Due to its high recall, the XGBoost model would suit the early drought warning systems. The feature importance analysis indicates that NDVI-based characteristics are the most important ones, which explains why they are important in monitoring drought.

Also, the study shows that temporal dependencies increase model performance by modeling delayed vegetation responses to rainfall. The proposed approach is capable of predicting drought in a scalable and efficient manner without relying on complex deep learning models.The further studies can focus on improving the accuracy of the model by focusing better on the imbalance of classes and incorporating more environmental factors.


## ACKNOWLEDGMENT

We are grateful to the PES University that it has granted the academic environment, resources and the opportunity to conduct this research. We would like to thank our mentors and faculty members who have guided us, provided constructive feedback and supported us throughout this work..


## REFERENCES


[1] Thanabalan, P., "Time-Series Analysis of MODIS (LST and NDVI) and TRMM Rainfall for Drought Assessment," Applied Geomatics, 2023.

[2] Bocchino, F., "Hierarchical Robust Combined Index for Agricultural Drought Detection," IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2026.

[3] Sruthi, S., "Agricultural Drought Analysis Using NDVI and Land Surface Temperature," Aquatic Procedia, 2015.

[4] Venkadesh, S. (2019). "Assessment of Agricultural Drought Using MODIS NDVI," International Journal of Current Microbiology and Applied Sciences.

[5] Peters, A. J. "Drought Monitoring with NDVI-Based Standardized Vegetation Index," Photogrammetric Engineering & Remote Sensing, 2002.

[6] Kamble, M. V. (2010). "Drought Monitoring Over India Using NDVI," Mausam Journal.

[7] Wankhede, S., "Agricultural Drought Forecasting Utilizing NDVI, VCI, and ANN," 2019.

[8] Sudhakara, B. "Drought Detection Using Spatio-Temporal Graph Neural Networks," IEEE Conference on Emerging Technology (INCET), 2025.

[9] Hoek, M. V., "Early Drought Detection by Spectral Analysis of Satellite Time Series of Precipitation and NDVI," Remote Sensing, 2016.

[10] General NDVI Studies (Multiple), "Use of Satellite NDVI Data for Vegetation Index Analysis," Various.

NDVI is widely accepted as a drought indicator, but NDVI alone is insufficient for accurate prediction.
[11] Rainfall-NDVI Integration Studies, "Correlation Analysis of Rainfall and NDVI Datasets for Drought Detection," Various. Combining rainfall and NDVI improves drought detection; however, these studies do not include machine learning models.
[12] ML-based Environmental Studies, "Application of Random Forest and Gradient Boosting on Multi-source Environmental

Data," Recent. These methods effectively capture non-linear relationships but often lack temporal lag feature integration.

[13] Time-Series Drought Studies, "Lag-based Modeling Using Climate Time-Series Data," Various. These studies show delayed vegetation response to rainfall but are not combined with machine learning models.

[14] Traditional Drought Models, "Statistical Analysis Using Rainfall and SPI Indices," Various. SPI provides simple and interpretable drought indicators but cannot capture complex non-linear interactions.

[15] Ensemble Learning Studies, "Use of XGBoost and Random Forest on Multi-feature Datasets for Classification," Recent. These models achieve high performance but struggle with class imbalance and noisy data.

[16] Vicente-Serrano, S. M., Beguería, S., and López-Moreno, J. I., "A Multiscalar Drought Index Sensitive to Global Warming: The Standardized Precipitation Evapotranspiration Index (SPEI)," *Journal of Climate*, vol. 23, no. 7, pp. 1696–1718, 2010.

[17] Mishra, A. K., and Singh, V. P., "A Review of Drought Concepts," *Journal of Hydrology*, vol. 391, no. 1–2, pp. 202–216, 2010.

[18] Kogan, F. N., "Global Drought Watch from Space," *Bulletin of the American Meteorological Society*, vol. 78, no. 4, pp. 621– 636, 1997.

[19] Rhee, J., Im, J., and Carbone, G. J., "Monitoring Agricultural Drought for Arid and Humid Regions Using Multi-sensor Remote Sensing Data," *Remote Sensing of Environment*, vol. 114, no. 12, pp. 2875–2887, 2010.

[20] Park, S., Im, J., Park, S., and Rhee, J., "Drought Assessment Using Machine Learning and Remote Sensing Data," *GIScience & Remote Sensing*, vol. 53, no. 4, pp. 516–534, 2016.

[21] AghaKouchak, A., "A Multivariate Approach for Persistence-Based Drought Prediction: Application to the United States," *Journal of Hydrology*, vol. 526, pp. 127–135, 2015.

AI-Based Drought Prediction Using Multi-Source Satellite Data and Machine Learning Models

ORIGINALITY REPORT

| 10% | 4% | 8% | 4% |
|---|---|---|---|
| SIMILARITY INDEX | INTERNET SOURCES | PUBLICATIONS | STUDENT PAPERS |

PRIMARY SOURCES

| | | |
|---|---|---|
| 1 | Paolo Ferro, Harinadh Vemanaboina, Chander Prakash. "Computational Techniques and Smart Manufacturing", CRC Press, 2026<br>Publication | 1% |
| 2 | K. V. Sambasivarao, Anasuya Sesha Roopa Devi Bhima. "Artificial Intelligence, Computational Intelligence, and Inclusive Technologies - Proceedings of International Conference on Artificial Intelligence, Computational Intelligence, and Inclusive Technologies (ICRAIC2IT – 2025)", CRC Press, 2026<br>Publication | 1% |
| 3 | www.mdpi.com<br>Internet Source | 1% |
| 4 | Submitted to Brunel University<br>Student Paper | 1% |
| 5 | Malaika Vijay, Amith Kashyap, Aushim Nagarkatti, Shruti Mohanty, Rajasekar Mohan, Niranjana Krupa. "Extreme Gradient Boosting Classification of Motor Imagery using Common Spatial Patterns", 2020 IEEE 17th India Council International Conference (INDICON), 2020<br>Publication | 1% |
| 6 | jicrcr.com<br>Internet Source | 1% |
| 7 | Submitted to University of Northumbria at Newcastle | 1% |

Student Paper

| | | |
|---|---|---|
| 8 | Jayashree R, Vinay S K. "A Jaccards Similarity Score Based Methodology for Kannada Text Document Summarization", 2020 International Conference on Advances in Computing, Communication & Materials (ICACCM), 2020<br>Publication | <1 % |
| 9 | Submitted to University of Wales Institute, Cardiff<br>Student Paper | <1 % |
| 10 | Submitted to The University of the West of Scotland<br>Student Paper | <1 % |
| 11 | www.nature.com<br>Internet Source | <1 % |
| 12 | balticpapers.com<br>Internet Source | <1 % |
| 13 | www.ijraset.com<br>Internet Source | <1 % |
| 14 | www.nass.usda.gov<br>Internet Source | <1 % |
| 15 | assets-eu.researchsquare.com<br>Internet Source | <1 % |
| 16 | dspace.univ-tlemcen.dz<br>Internet Source | <1 % |
| 17 | iieta.org<br>Internet Source | <1 % |
| 18 | worldwidescience.org<br>Internet Source | <1 % |
| 19 | Fatemeh Ghasempour, Sevim Seda Yamaç, Aliihsan Sekertekin, Muzaffer Can Iban, Senol Hakan Kutoglu. "Spatiotemporal Agricultural Drought Assessment and Mapping Its | <1 % |

Vulnerability in a Semi-Arid Region Exhibiting Aridification Trends", Agriculture, 2025
Publication

---

| | | |
|---|---|---|
| 20 | Quilty, John, Jan Adamowski, Bahaa Khalil, and Maheswaran Rathinasamy. "Bootstrap rank-ordered conditional mutual information (broCMI)-A nonlinear input variable selection method for water resources modeling", Water Resources Research, 2016.<br>Publication | <1 % |
| 21 | Kumwenda, Vitumbiko Blessings. "Predicting Primary School Learner Dropout in Malawi Using Machine Learning-Based Predictive Scorecard.", Malawi University of Science and Technology<br>Publication | <1 % |
| 22 | Mustafa, Meher Nigar. "Leveraging Data Science to Address Mental Health Challenges: Insights into Social Media, Academic Stress, and Faculty Support Among University Students.", Lamar University - Beaumont<br>Publication | <1 % |

| | | | |
|---|---|---|---|
| Exclude quotes | Off | Exclude matches | Off |
| Exclude bibliography | On | | |